1. Staging of Chronic Kidney Disease based on blood creatinine and SDMA concentrations

Staging is undertaken following diagnosis of chronic kidney disease (CKD) to facilitate appropriate treatment and monitoring of the canine or feline patient. Staging is based initially on fasting blood creatinine or fasting blood SDMA concentration or (preferably) both assessed on at least two occasions in a hydrated, stable patient. The dog or cat is then substaged based on proteinuria and blood pressure.

Using these criteria, some empirical recommendations can be made about the type of treatment it would be logical to use for these cases. In addition, predictions based on clinical experience might be made about the likely response to treatment.

IRIS Staging of CKD (Modified 2026)

Stage	1		2		3		4	
	Dogs	**Cats**	**Dogs**	**Cats**	**Dogs**	**Cats**	**Dogs**	**Cats**
Blood creatinine* µmol/l	<125	<140	125 –250	140 – 250	251 – 440	251 – 440	>440	>440
Blood creatinine* mg/dl	<1.4	<1.6	1.4 – 2.8	1.6 – 2.8	2.9 – 5.0	2.9 – 5.0	>5.0	>5.0
SDMA µg/dl	<18	<18	18 – 35	18 – 25	36 – 54	26 – 38	>54	>38

Comments

Stage 1

Normal blood creatinine or normal or mild increase blood SDMA. Some other renal abnormality present (such as, inadequate urinary concentrating ability without identifiable non-renal cause (in cats not dogs), abnormal renal palpation or renal imaging findings, proteinuria of renal origin, abnormal renal biopsy results, increasing blood creatinine or SDMA concentrations in samples collected serially). Persistently elevated blood SDMA concentration (>14 µg/dl) may be used to diagnose early CKD.

Stage 2

Normal or mildly increased creatinine, mild renal azotemia (lower end of the range lies within reference ranges for creatinine for many laboratories, but the insensitivity of creatinine concentration as a screening test means that patients with creatinine values close to the upper reference limit often have excretory failure). Mildly increased SDMA. Clinical signs usually mild or absent.

Stage 3

Moderate renal azotemia. Many extrarenal signs may be present, but their extent and severity may vary. If signs are absent, the case could be considered as early Stage 3, while presence of many or marked systemic signs might justify classification as late Stage 3.

Stage 4

Increasing risk of systemic clinical signs and uremic crises.

* The blood creatinine concentrations apply to average size dogs – those of extreme size may vary.

The recommendations for SDMA are based on published literature which utilizes proprietary IDEXX technology for measuring SDMA.

At this time, it is not known if other assays will provide equivalent results.

Discrepancies between creatinine and SDMA

IRIS CKD staging is based on fasting blood creatinine concentration and blood SDMA concentration. SDMA is potentially a more sensitive marker that is less impacted by loss of lean body mass.

DOGS:

Stage and Treat as IRIS CKD Stage 2
If serum or plasma SDMA is persistently >18 µg/dl in a dog whose creatinine is **< 125 µmol/l [1.4 mg/dl]**
(IRIS CKD stage 1 based on creatinine).

Stage and Treat as IRIS CKD Stage 3
If serum or plasma SDMA is persistently >35 µg/dl in a dog whose creatinine is between **125 and 250 µmol/l [1.4 and 2.8 mg/dl]**
(IRIS CKD stage 2 based on creatinine).

Stage and Treat as IRIS CKD Stage 4
If serum or plasma SDMA is persistently >54 µg/dl in a dog whose creatinine is between **251 and 440 µmol/l [2.9 and 5.0 mg/dl]**
(IRIS CKD stage 3 based on creatinine).

CATS:

Stage and Treat as IRIS CKD Stage 2
If serum or plasma SDMA is persistently >18 µg/dl in a cat whose creatinine is **<140 µmol/l [1.6 mg/dl]**
(IRIS CKD stage 1 based on creatinine).

Stage and Treat as IRIS CKD Stage 3
If serum or plasma SDMA is persistently >25 µg/dl in a cat whose creatinine is between **141 and 250 µmol/l [1.6 and 2.8 mg/dl]**
(IRIS CKD stage 2 based on creatinine).

Stage and Treat as IRIS CKD Stage 4
If serum or plasma SDMA is persistently >38 µg/dl in a cat whose creatinine is between **251 and 440 µmol/l [2.9 and 5.0 mg/dl]**
(IRIS CKD stage 3 based on creatinine).

SDMA assays are offered by a number of laboratories throughout the world. The methodology used has not yet been standardized and the recommendations made above are based on the proprietary methodology offered by IDEXX Laboratories.

These recommendations are based on current state of knowledge where SDMA appears to be a more sensitive indicator of early-stage CKD in the dog and cat.

Data are starting to emerge on conditions where SDMA may be elevated without a reduction in GFR, where serum creatinine values are unaffected. Lymphoma in dogs and cats is one such example. This illustrates the value of having two biomarkers which are surrogate markers for GFR which can be interpreted together.

Breed and size associated effects on serum creatinine and SDMA

Healthy Birman cats and greyhound dogs have been found to have higher serum SDMA values than other breeds. Both these breeds also have higher serum creatinine values, taking some healthy individuals (up to 20% of Birman cats) outside laboratory reference intervals.

2a. Substaging by Proteinuria

The goal is to identify renal proteinuria having ruled out post-renal and pre-renal causes.

Standard urine dipsticks can be insensitive therefore practitioners should consider using a more reliable quantitative test such as the urine protein to creatinine ratio (UP/C) or a species-specific albuminuria assay.

The UP/C should be measured in all dogs and cats with CKD, provided there is no evidence of urinary tract inflammation or hemorrhage, and the routine measurement of plasma proteins has ruled out dysproteinemias. Ideally sub-staging should be done on the basis of at least two urine samples collected over a period of at least 2 weeks.

UP/C value		Substage
Dogs	**Cats**	
<0.2	<0.2	Non-proteinuric
0.2 to 0.5	0.2 to 0.4	Borderline proteinuric
>0.5	>0.4	Proteinuric

Canine and feline patients that are persistently borderline proteinuric should be re-evaluated within 2 months and re-classified as appropriate.

The significance of borderline proteinuria in predicting future renal health is not completely understood. IRIS's recommendation is to continue to monitor this level of proteinuria in dogs. Veterinarians might offer treatment for cats with persistent borderline proteinuria considering the association with progressive kidney disease (see treatment guidelines).

Response to any treatment given to reduce glomerular hypertension, filtration pressure, and proteinuria, should be monitored at intervals using UP/C.

2b. Substaging by Blood pressure

Canine and feline patients should be acclimatized to the measurement conditions and multiple measurements taken. The final classification should rely upon multiple systolic blood pressure determinations, preferably done during repeated patient visits to the clinic on separate days, but acceptable if during the same visit with at least 2 hours separating determinations. Patients are substaged by systolic blood pressure according to the degree of risk of target organ damage, and whether there is evidence of target organ damage or complications.

For most dogs and all cats, the IRIS blood pressure substages are as follows:

Systolic Blood Pressure mmHg	Blood Pressure Substage	Risk of Future Target Organ Damage
<140	**Normotensive**	Minimal
140 – 159	**Prehypertensive**	Low
160 – 179	**Hypertensive**	Moderate
≥ 180	**Severely hypertensive**	High

However, some breeds of dog, particularly sight hounds, tend to have higher blood pressure than other breeds. It is preferable to use breed-specific reference ranges if available. The classification of risk of future target organ damage in “high-pressure breeds” might be adjusted as follows:

Minimal risk	systolic pressure <10 mm Hg above the breed-specific reference interval
Low risk	systolic pressure 10-20 mm Hg above the breed-specific reference interval
Moderate risk	systolic pressure 20-40 mm Hg above the breed-specific reference interval
High risk	systolic pressure >40 mm Hg above the breed-specific reference interval.

As with proteinuria, in the absence of evidence of existing target organ damage, demonstration of persistence of blood pressure readings within a particular category is important. ‘Persistence’ of increased systolic blood pressure should be judged on multiple measurements made at different clinic visits (within 1-2 weeks). The higher the systolic blood pressure, the shorter the time frame over which persistence should be judged.

3. Revision of staging and substaging after treatment

The stage and substages assigned to the patient should be revised appropriately as changes occur. For example, a substantial increase in blood creatinine or SDMA concentration might warrant reassignment to a higher stage to reflect the new situation.

Similarly, if antihypertensive (or antiproteinuric) treatment has been instituted, the patient's classification on re-evaluation should be adjusted if necessary to reflect the new blood pressure (or UP/C) rather than the original status, with the addition of an indication that the current classification is affected by treatment.

The following two examples illustrate the process of revision, where 'treating' is used as an indicator of ongoing treatment.

Example 1

Euvolemic cat with stable renal function
Creatinine 200 µmol/l [2.3 mg/dl]
SDMA 22 µg/dl
UP/C 0.32
Systolic blood pressure 200 mm Hg
Classification – *IRIS CKD Stage 2, borderline proteinuric, severely hypertensive.*

Same cat after antihypertensive treatment
Creatinine 220 µmol/l [2.5 mg/dl]
SDMA 24 µg/dl
UP/C 0.12
Systolic blood pressure 155 mm Hg
New classification – *IRIS CKD Stage 2, non-pro-teinuric, prehypertensive (treating).*

Example 2

Euvolemic dog with stable renal function
Creatinine 230 µmol/l [2.6 mg/dl]
SDMA 39 µg/dl
UP/C 0.8
Systolic blood pressure 155 mm Hg
Classification – *IRIS CKD Stage 3, proteinuric, prehypertensive*

Note: As described above in the **Discrepancies between creatinine and SDMA** *section, if blood SDMA is persistently >35 µg/dl in a canine patient whose blood creatinine is between 125 and 250 µmol/l (1.4 and 2.8 mg/dl; IRIS CKD stage 2 based on creatinine), this dog should be staged and treated as an IRIS CKD Stage 3 patient.*

Same dog after antiproteinuric treatment
Creatinine 240 µmol/l [2.7 mg/dl]
SDMA 42 µg/dl
UP/C 0.4
Systolic blood pressure 155 mm Hg
New classification – IRIS CKD Stage 3, borderline proteinuric (treating), prehypertensive.

All treatments for chronic kidney disease (CKD) need to be tailored to the individual patient. The following recommendations are useful starting points for the majority of dogs at each stage. Serial monitoring of these patients is ideal and treatment should be modified according to the response to treatment. Note that staging of disease is undertaken following diagnosis of CKD – an increased blood creatinine or symmetric dimethylarginine (SDMA) concentration alone is not diagnostic of CKD.

Treatment recommendations fall into two broad categories, namely:

i. Those that slow progression of CKD and thereby preserve remaining kidney function for longer

ii. Those that seek to improve the quality of life of the dog, reducing signs of CKD

In general, there are few clinical extra-renal signs at the early stages of CKD (Stages 1 and 2) and the therapeutic emphasis is on slowing progression.

From Stage 3 onwards, extra-renal signs become more common and more severe. By Stage 4, treatments that are directed towards mollifying clinical signs and improve quality of life assume greater importance, and become more relevant than those designed to slow progression of CKD. Some of the treatment recommendations are not authorized for use in all geographical regions and some may not be authorized for use in dogs.
Such recommended dose rates are therefore empirical. It is the treating veterinarian's duty to make a risk:benefit assessment for each dog prior to administering any treatment.

Treatment recommendations for Dogs with Chronic Kidney Disease

Stage 1 canine patients:

1. Discontinue all potentially nephrotoxic drug if possible.

2. Identify and treat any pre-renal or post-renal abnormalities.

3. Rule out any treatable conditions like pyelonephritis and ureteral obstruction with radiographs and/or ultrasonography.

4. Measure blood pressure and urine protein to creatinine ratio (UP/C).

Management of dehydration:

In these patients, urine concentrating ability may be somewhat impaired and therefore ensure:

1. They have fresh water available at all times for drinking

2. If patient becomes ill for any reason that leads to fluid losses, correct clinical dehydration with isotonic polyionic replacement fluid solutions (e.g. lactated Ringer's) IV or SQ, promptly as needed

Systemic hypertension:

The blood pressure above which progressive kidney injury may be induced is unknown. Our goal is to reduce systolic blood pressure to <160 mm Hg and minimize the risk of extra-renal target organ damage (CNS, retinal, cardiac problems/damage). If there is no evidence of such damage but systolic blood pressure persistently exceeds 160 mm Hg, treatment should be instituted.

1. Persistence of increase in systolic blood pressure should be judged on multiple measurements made over the following time-scales in these blood pressure substages:

2. Hypertensive (moderate risk of future target organ damage) – systolic blood pressure 160 to 179 mm Hg, persistence demonstrated over 2 to 4 weeks

3. Severely hypertensive (high risk of future target organ damage) – systolic blood pressure >180 mm Hg persistence demonstrated over 1 to 2 weeks

4. If evidence of target organ damage exists, dogs should be treated without the need to demonstrate persistently increased systolic blood pressure. Reducing blood pressure is a long term aim when managing the patient with CKD and a gradual and sustained reduction should be the goal, avoiding any sudden or severe decreases leading to hypotension.

5. It is recognized that some breeds (such as sight hounds) tend to have higher blood pressure (see Appendix 1) and that this may influence interpretation.

A logical stepwise approach to managing hypertension is as follows:

1. Dietary sodium (Na) reduction – there is no evidence that lowering dietary Na will reduce blood pressure. If dietary Na reduction is attempted, it should be accomplished gradually and in combination with pharmacological therapy.

2. Angiotensin converting enzyme inhibitor (ACEI, such as benazepril) therapy at standard dose rate.

3. Double the dose of ACEI (in some dogs, increasing the dose may improve the antihypertensive effect).

4. Combine ACEI and calcium channel blocker (CCB, such as amlodipine) treatment, especially if severely hypertensive.

5. Combine ACEI and CCB with angiotensin receptor blocker (ARB, such as telmisartan) and/or hydralazine if additional treatment is required.

Note: Take care not to introduce ACEI/CCB with or without ARB to unstable dehydrated dogs as glomerular filtration rate may drop precipitously if these drugs are introduced before the patient is adequately hydrated. The risk benefit analysis of combining ACEI with ARBs needs to be made on an individual dog basis and careful monitoring is required to ensure any deterioration in kidney function is detected.

Monitoring response to antihypertensive treatment:

1. Hypertensive dogs normally require lifelong therapy and frequently require adjustments in treatment. Serial monitoring is essential. After stabilization, monitoring should occur at least every 3 months.

2. Systolic blood pressure <120 mm Hg and/or clinical signs such as weakness or tachycardia indicate hypotension, which is to be avoided.

3. Blood creatinine concentration – reducing blood pressure may lead to small and persistent increases in creatinine concentration (< 45 µmol/l or 0.5 mg/dl increase) and/or SDMA (< 2 µg/dl), but a marked increase suggests an adverse drug effect. Progressively increasing concentrations indicate progressive kidney damage/disease.

Proteinuria:

Dogs in Stage 1 with UP/C > 0.5 should be investigated for disease processes leading to proteinuria. Those with confirmed proteinuria and treated with antiproteinuric measures require close monitoring. The goal of treatment is nuanced. Treatment should aim to reduce proteinuria to the lowest UPC possible without doing harm. Those with borderline proteinuria (0.2-0.5) require close monitoring.

1. Look for any concurrent associated disease process that may be treated/corrected.

2. Consider kidney biopsy (for dogs in stages 1 to 3, not stage 4) as a means of identifying underlying disease (see Appendix 2 and/or consult experts if unsure of indications for kidney biopsy).

3. Administer an angiotensin receptor blocker (ARB) and feed a clinical kidney diet.

4. Combination of an ACEI and diet with an angiotensin receptor blocker (ARB) if proteinuria is not controlled should be done judiciously and cautiously, ideally under consultation with a veterinary nephrologist (see note a. below).

5. Consider clopidogrel (1.1-3 mg/kg orally every 24 hours). If clopidogrel is not available, low-dose acetylsalicylic acid (2 to 5 mg/kg once daily) is an acceptable alternative (see note b. below).

6. Monitor response to treatment / progression of disease:
   - Stable blood creatinine concentration, decreasing UP/C and/or increasing serum albumin (if previously low) = good response.
   - A UPC of < 0.5 is not achievable for many dogs with primary glomerular disease. For these dogs the goal should be a 50% reduction in UPC from baseline
   - Serially increasing blood creatinine concentration and/or increasing UP/C = disease is progressing.

Ordinarily therapy will be maintained lifelong unless the underlying disease has been resolved in which case dose reduction whilst monitoring UP/C might be considered.

Note:

a. ACEI or ARB use is contraindicated in any dog that is clinically dehydrated and/or is showing signs of hypovolemia. Correct dehydration before using these drugs otherwise glomerular filtration rate may drop precipitously. The risk benefit analysis of combining ACEI with ARBs needs to be made on an individual dog basis and careful monitoring is required to ensure any deterioration in kidney function is detected. Further detailed recommendations for diagnosis and management of glomerular disease in dogs can be found in the IRIS Consensus Statements published in Journal of Veterinary Internal Medicine in 2013 (supplement to volume 27, https://onlinelibrary.wiley.com/toc/19391676/2013/27/s1).

b. Dogs with PLN are at risk of thrombosis however it is not possible to predict the risk for thrombosis in the individual patient as serum albumin, antithrombin and degree of proteinuria are poorly associated with thrombotic risk. Tests such as thromboelastography, thrombin generation and markers of activated clotting can be used to document hypercoagulability, however, identification of a hypercoagulable state has not been shown to correlate to risk of developing thrombosis. Prothrombin and partial thrombloplastin times (PT, PTT) also cannot be used to predict thrombotic risk.

   Antithrombotic therapy is indicated in dogs with PLN (CURATIVE Guidelines DOI: 10.1111/vec.12801). Clopidogrel (1-4mg/kg orally once daily) administration may be more effective than low dose acetylsalicylic acid (1-5mg/kg orally once daily) for thromboprophylaxis.

Stage 2 Canine patients:

All of the above listed for Stage 1 (listed here again for convenience), plus any additional steps indicated below.

1. Discontinue all potentially nephrotoxic drugs if possible.
2. Identify and treat any pre-renal or post-renal abnormalities.
3. Rule out any treatable conditions like pyelonephritis and ureteral obstruction with radiographs and/or ultrasonography.
4. Measure blood pressure and urine protein to creatinine ratio (UP/C).
5. Consider feeding a clinical renal diet: this may be accomplished more easily early in the course of CKD, before inappetence develops.

Management of dehydration:

These canine patients have decreased urine concentrating ability and therefore ensure:

1. They have fresh water available at all times for drinking.
2. If they become ill for any reason leading to fluid losses, correct clinical dehydration/hypovolemia with isotonic, polyionic replacement fluid solutions (e.g., lactated Ringer's) IV or SQ promptly as needed

Systemic hypertension:

The blood pressure above which progressive kidney injury may be induced is unknown. Our goal is to reduce systolic blood pressure to <160 mm Hg and minimize the risk of extra-renal target organ damage (CNS, retinal, cardiac problems/damage). If there is no evidence of this but systolic blood pressure persistently exceeds 160 mm Hg, increasing the risk of such damage, treatment should be instituted.

Persistence of increase in systolic blood pressure should be judged on multiple measurements made over the following time-scales in these blood pressure substages:

1. Hypertensive (moderate risk of future target organ damage) – systolic blood pressure 160 to 179 mm Hg – persistence demonstrated over 2 to 4 weeks
2. Severely hypertensive (high risk of future target organ damage) – systolic blood pressure >180 mm Hg persistence demonstrated over 1 to 2 weeks
3. If evidence of target organ damage exists, dogs should be treated without the need to demonstrate persistently increased systolic blood pressure. Reducing blood pressure is a long term aim in a patient with CKD and a gradual and sustained reduction should be the goal, avoiding any sudden decreases or hypotension.

A logical stepwise approach to managin hypertension is as follows:

1. Dietary sodium (Na) reduction – there is no evidence that lowering dietary Na will reduce blood pressure. If dietary Na reduction is attempted, it should be accomplished gradually and in combination with pharmacological therapy.

2. Angiotensin converting enzyme inhibitor (ACEI, such as benazepril) therapy at standard dose rate.

3. Double the dose of ACEI (in some patients, increasing the dose may improve the antihypertensive effect).

4. Combine ACEI and calcium channel blocker (CCB, such as amlodipine) treatment, especially if severely hypertensive.

5. Combine ACEI and CCB with angiotensin blocker (ARB, such as telmisartan) and/or hydralazine if additional treatment is required.

Note: Take care not to introduce ACEI/CCB with or without ARB treatment to unstable dehydrated dogs as glomerular filtration rate may drop precipitously if these drugs are introduced before the patient is adequately hydrated. The risk benefit analysis of combining ACEI with ARBs needs to be made on an individual dog basis and careful monitoring is required to ensure any deterioration in kidney function that is detected.

Monitoring response to antihypertensive treatment:

1. Hypertensive dogs normally require lifelong therapy and frequently require adjustments in treatment dosages. Serial monitoring is essential. After stabilization, monitoring should occur at least every 3 months.

2. Systolic blood pressure <120 mm Hg and/or clinical signs such as weakness or tachycardia indicate hypotension, which is to be avoided.

3. Blood creatinine concentration – reducing blood pressure may lead to small and persistent increases in creatinine (< 45 µmol/l or 0.5 mg/dl increase) and/or SDMA (< 2 µg/dl), but a marked increase suggests an adverse drug effect. Progressively increasing concentrations indicate progressive kidney damage/disease.

Proteinuria:

Dogs in Stage 2 with UP/C >0.5 should be investigated for the disease processes leading to proteinuria. Those with confirmed and persistent renal proteinuria should be treated with anti-proteinuric measures. The goal of treatment is nuanced. The objective of treatment is to reduce proteinuria to the lowest UPC achievable without causing harm.
Those with borderline proteinuria (0.2 to 0.5) require close monitoring.

1. Look for any concurrent associated disease process that may be treated/corrected.

2. Consider kidney biopsy (for dogs in stages 1 to 3, not stage 4) as a means of identifying underlying disease (see Appendix 2 and/or consult experts if unsure of indications for kidney biopsy).

3. Administer an angiotensin receptor blocker (ARB) and feed a clinical renal diet.

4. Combination of an ACEI and diet with an ARB if proteinuria is not controlled should be done judiciously and cautiously, ideally under consultation with a veterinary nephrologist (see note 1 below).

5. Consider clopidogrel (1.1-3 mg/kg orally every 24 hours). If clopidogrel is not available, low-dose acetylsalicylic acid (2 to 5 mg/kg once daily) is an acceptable alternative (see note 2 below).

6. Monitor response to treatment / progression of disease:

- Stable blood creatinine concentration, decreasing UP/C and/or increasing serum albumin (if previously low) = good response.
- A UPC of < 0.5 is not achievable for many dogs with primary glomerular disease. For these dogs the goal should be a 50% reduction in UPC from baseline.

- Serially increasing blood creatinine concentrations and/or increasing UP/C and/or increasing serum albumin (if previously low) = disease is progressing.

Ordinarily therapy will be maintained lifelong unless the underlying disease has been resolved in which case dose reduction whilst monitoring UP/C might be considered.

Note:

a. ACEI or ARB use is contraindicated in any animal that is clinically dehydrated and/or showing signs of hypovolemia. Correct dehydration before using these drugs otherwise glomerular filtration rate may drop precipitously. The risk benefit analysis of combining ACEI with ARBs needs to be made on an individual dog basis and careful monitoring is required to ensure any deterioration in kidney function is detected. Further detailed recommendations for diagnosis and management of glomerular disease in dogs can be found in the IRIS Consensus Statements published in Journal of Veterinary Internal Medicine in 2013 (supplement to volume 27, https://onlinelibrary. wiley.com/toc/19391676/2013/27/s1).

b. Dogs with PLN are at risk of thrombosis however it is not possible to predict the risk for thrombosis in the individual patient as serum albumin, antithrombin and degree of proteinuria are poorly associated with thrombotic risk. Tests such as thromboelastography, thrombin generation and markers of activated clotting can be used to document hypercoagulability, however, identification of a hypercoagulable state has not been shown to correlate to risk of developing thrombosis. Prothrombin and partial thrombloplastin times (PT, PTT) also cannot be used to predict thrombotic risk.

   Antithrombotic therapy is indicated in dogs with PLN (CURATIVE Guidelines DOI: 10.1111/vec.12801). Clopidogrel (1-4mg/kg orally once daily) administration may be more effective than low dose acetylsalicylic acid (1-5mg/kg orally once daily) for thromboprophylaxis

Reduction of phosphate intake:

Evidence suggests that chronic reduction of phosphate intake to maintain a plasma phosphate concentration below 1.5 mmol/l (but not less than 0.9 mmol/l; < 4.6 mg/dl but > 2.7 mg/dl) is beneficial to patients with CKD.

The following measures can be introduced sequentially in an attempt to achieve this:

1. Dietary phosphate restriction (i.e., clinical kidney diet therapy).

2. If plasma phosphate concentration remains above 1.5 mmol/l (4.6 mg/dl) after dietary restriction, give enteric phosphate binders (such as aluminum hydroxide, aluminum carbonate, calcium carbonate, calcium acetate, lanthanum carbonate, ferric citrate) to effect, starting at 30-60 mg/kg/day in divided doses to be mixed with each meal (mixed with the food). The dose required will vary according to the amount of phosphate being fed and the stage of kidney disease. Treatment with phosphate binders should be to effect (as outlined above), with signs of toxicity limiting the upper dose rate possible in a given patient. Monitor serum calcium and phosphate concentrations every 4-6 weeks until stable and then every 12 weeks. Microcytosis and/or generalized muscle weakness suggests aluminum toxicity if using an aluminum containing binder – switch to another form of phosphate binder should this occur. Hypercalcemia should be avoided – combinations of aluminum and calcium containing phosphate binders may be necessary in some cases.

Additional recommendations for Stage 2 patients aimed at improving quality of life:

Anemia:

1. Anemia is a common problem in dogs and cats with chronic kidney disease. Observational studies have identified an association between anemia and survival in cats with CKD. While interventional studies have not been performed to determine the optimal treatment targets for hematocrit levels in dogs or cats, observations suggest that treating anemia associated with chronic kidney disease can improve the quality of life for pets. Treatment options for anemia due to CKD includes darbepoetin, HIF-PH inhibitors such as molidustat and blood transfusions.

2. Anemia can have multiple causes and animals should also be evaluated and treated for contributing factors, such as infections or excessive blood sampling.

    I. Dogs with anemia due to CKD with a hematocrit of < 30% should be treated.

    II. Dogs with persistent anemia due to CKD with a hematocrit between 30 and 35% should be treated.

3. In a euhydrated dog with Stage 2 CKD, anemia of <30 % is less likely to be attributable to CKD and the patient should be assessed for other disease processes. If further workup does not reveal any other underlying cause, then the anemia is most likely attributable to CKD and should be treated.

Gastrointestinal signs:

1. Treat vomiting and suspected nausea with anti-emetics such as maropitant or ondansetron.

2. Treat decreased appetite/weight and/or muscle loss with appetite stimulants such as capromorelin or mirtazapine.

3. If muscle loss is marked, consider staging based on serum SDMA concentration rather than creatinine and following treatment recommendations for SDMA stage (see below)

4. Consider intermittent omeprazole in dogs with suspected gastric bleeding (eg melena, iron deficiency) or vomiting-induced esophagitis.

Where there is a discrepancy between creatinine and SDMA:

If serum or plasma SDMA is > 35 µg/dl in a patient whose creatinine is between 1.4 and 2.8 mg/dl (IRIS CKD stage 2 based on blood creatinine), this patient should be staged and treated as an IRIS.

Stage 3 Canine patients:

The range of presentations for dogs in Stage 3 is wide, from no clinical signs to quite marked extra-renal signs. The treatments mentioned so far for Stages 1 and 2 aimed at slowing progression of CKD also apply in Stage 3 and may be all that is required for dogs with no or mild clinical signs. But treatments designed to improve the quality of life are more important as extra-renal signs become increasingly evident. These include treatments to address dehydration, nausea and vomiting, anemia and acidosis.

Treatment recommendations include all of the above listed for Stage 1 and 2 (listed again here for convenience) plus any additional steps indicated below.

1. Discontinue all potentially nephrotoxic drugs if possible.

2. Identify and treat any pre-renal or post-renal abnormalities.

3. Rule out any treatable conditions like pyelonephritis and ureteral obstruction with radiographs and/or ultrasonography.

4. Measure blood pressure and urine protein to creatinine ratio (UP/C).

5. Feed a clinical kidney diet.

Management of dehydration:

These patients have decreased urine concentrating ability and therefore

1. Correct clinical dehydration/hypovolemia with isotonic, polyionic replacement fluid solutions (e.g., lactated Ringer's) IV or SQ promptly as needed.

2. Have fresh water available at all times for drinking

3. In addition, some of these dogs may require maintenance fluids administered routinely to maintain hydration (see below)

Systemic hypertension:

The blood pressure above which progressive kidney injury may be induced is unknown. Our goal is to reduce systolic blood pressure to <160 mm Hg and minimize the risk of extra-renal target organ damage (CNS, retinal, cardiac problems/damage). If there is no evidence of this but systolic blood pressure persistently exceeds 160 mm Hg, increasing the risk of such damage, treatment should be instituted.

Persistence of increased systolic blood pressure should be judged on multiple measurements made over the following time-scales in these blood pressure substages:

1. Hypertensive (moderate risk of future target organ damage) – systolic blood pressure 160 to 179 mm Hg, persistence demonstrated over 2 to 4 weeks
2. Severely hypertensive (high risk of future target organ damage – systolic blood pressure > 180 mm Hg measured, persistence demonstrated over 1 to 2 weeks

If evidence of target organ damage exists, dogs should be treated without the need to demonstrate persistently increased systolic blood pressure. Reducing blood pressure is a long term aim when managing the patient with CKD and a gradual and sustained reduction should be the goal, avoiding any sudden or severe decreases leading to hypotension.
Some breeds (such as sight hounds) tend to have higher blood pressure (see Appendix) and that this may influence interpretation.

A logical stepwise approach to managing hypertension is as follows:

1. Dietary sodium (Na) reduction - there is no evidence that lowering dietary Na will reduce blood pressure. If dietary Na reduction is attempted, it should be accomplished gradually and in combination with pharmacological therapy.

2. Angiotensin converting enzyme inhibitor (ACEI, such as benazepril) therapy at standard dose rate.

3. Double the dose of ACEI (in some patients, increasing the dose may improve the anti-hypertensive effect).

4. Combine ACEI and calcium channel blocker (CCB, such as amlodipine) treatment, especially if severely hypertensive.

5. Combine ACEI and CCB with angiotensin receptor blocker (ARB, such as telmisartan) and/or hydralazine if additional treatment is required.

Note: Take care not to introduce ACEI/CCB with or without ARB treatment to unstable dehydrated dogs as glomerular filtration rate may drop precipitously if these drugs are introduced before the patient is adequately hydrated. The risk benefit analysis of combining ACEI with ARBs needs to be made on an individual dog basis and careful monitoring is required to ensure any deterioration in kidney function that is detected. The risk is higher in CKD stages 3 and 4.

Monitoring response to antihypertensive treatment:

Hypertensive dogs normally require lifelong therapy and may require treatment adjustments. Serial monitoring is essential. After stabilization, monitoring should occur at least every 3 months.

Systolic blood pressure <120 mm Hg and/or clinical signs such as weakness or tachycardia indicate hypotension, which is to be avoided.

Blood creatinine concentration – reducing blood pressure may lead to small and persistent increases in blood creatinine (<45 µmol/l or 0.5 mg/dl increase) and/or SDMA (< 2 µg/dl), but a marked increase suggests an adverse drug effect. Progressively increasing creatinine concentrations indicate progressive kidney damage/ disease.

Proteinuria:

Dogs in Stage 3 with urine protein to creatinine ratio (UP/C) >0.5 should be investigated for disease processes leading to proteinuria. Those with confirmed and persistent renal proteinuria should be treated with anti-proteinuric measures. The goal of treatment is nuanced. Treatment should reduce proteinuria to the lowest possible UPC, without doing harm.

Those with borderline proteinuria (0.2 to 0.5) require close monitoring.

1. Look for any concurrent associated disease process that may be treated/corrected.

2. Consider kidney biopsy (for dogs in stages 1 to 3, not stage 4) as a means of identifying underlying disease (see Appendix 2 and/ or consult experts if unsure of indications for kidney biopsy).

3. Administer an angiotensin receptor blocker (ARB) and feed a clinical kidney diet.

4. Combination of an ACEI and diet with an ARB if proteinuria is not controlled, should be done judiciously and cautiously, ideally under consultation with a veterinary nephrologist (see note 1 below).

5. Consider clopidogrel (1.1- 3 mg/kg orally every 24 hours). If clopidogrel is not available, low-dose acetylsalicylic acid (2 to 5 mg/kg once daily) is an acceptable alternative (see note 2 below).

6. Monitor response to treatment / progression of

disease:

- Stable blood creatinine concentration, decreasing UP/C and/or increasing serum albumin (if previously low) = good response.

  A UPC of < 0.5 is not achievable for many dogs with primary glomerular disease. For these dogs the goal should be a 50% reduction in UPC from baseline

- Serially increasing blood creatinine concentration and/or increasing UP/C = disease is progressing.

Ordinarily therapy will be maintained lifelong unless the underlying disease has been resolved in which case reduction whilst monitoring UP/C might be considered.

Note:

a. ACEI or ARB use is contraindicated in any animal that is clinically dehydrated and/or is showing signs of hypovolemia. Correct dehydration before using these drugs otherwise glomerular filtration rate may drop precipitously. The risk benefit analysis of combining ACEI with ARBs needs to be made on an individual dog basis and careful monitoring is required to ensure any deterioration in kidney function is detected. The risk is higher in CKD stages 3 and 4.
Further detailed recommendations for diagnosis and management of glomerular disease in dogs can be found in the IRIS Consensus Statements published in Journal of Veterinary Internal Medicine in 2013 (supplement to volume 27, https://onlinelibrary.wiley.com/toc/19391676/2013/27/s1).

b. Dogs with PLN are at risk of thrombosis however it is not possible to predict the risk for thrombosis in the individual patient as serum albumin, antithrombin and degree of proteinuria are poorly associated with thrombotic risk. Tests such as thromboelastography, thrombin generation and markers of activated clotting can be used to document hypercoagulability, however, identification of a hypercoagulable state has not been shown to correlate to risk of developing thrombosis. Prothrombin and partial thrombloplastin times (PT, PTT) also cannot be used to predict thrombotic risk.

Antithrombotic therapy is indicated in dogs with PLN (CURATIVE Guidelines DOI: 10.1111/vec.12801). Clopidogrel (1-4mg/kg orally once daily) administration may be more effective than low dose acetylsalicylic acid (1-5mg/kg orally once daily) for thromboprophylaxis.

Reduction of phosphate intake:

Evidence suggests that chronic reduction ofphosphate intake to maintain a plasma phosphate concentration below 1.5 mmol/l (but not less than 0.9 mmol/l; <4.6 mg/dl but >2.7 mg/dl) is beneficial to patients with CKD.

A more realistic post-treatment target for dogs at Stage 3 is <1.6 mmol/l (5.0 mg/dl).

The following measures can be introduced sequentially in an attempt to achieve this:

1. Dietary phosphate restriction (i.e., clinical renal kidney therapy).

2. If plasma phosphate concentration remains above 1.6 mmol/l (5.0 mg/dl) after dietary restriction, give enteric phosphate binders (such as aluminum hydroxide, aluminum carbonate, calcium carbonate, calcium acetate, lanthanum carbonate, ferric citrate) to effect, starting at 30-60 mg/kg/day in divided doses to be mixed with each meal (mixed with the food). The dose required will vary according to the amount of phosphate being fed and the stage of kidney disease. Treatment with phosphate binders should be to effect (as outlined above), with signs of toxicity limiting the upper dose rate possible in a given patient. Monitor serum calcium and phosphate concentrations every 4-6 weeks until stable and then every 12 weeks. Microcytosis and/or generalized muscle weakness suggests aluminum toxicity if using an aluminum containing binder – switch to another form of phosphate binder should this occur. Hypercalcemia should be avoided – combinations of aluminum and calcium containing phosphate binders may be necessary in some cases.

Additional recommendations for Stage 3 patients aimed at improving quality of life:

1. Metabolic acidosis:

If metabolic acidosis exists (blood bicarbonate or total CO2 <18 mmol/l) once the patient is stabilized on the clinical kidney diet of choice, supplement with oral sodium bicarbonate (or potassium citrate if hypokalemic) to maintain blood bicarbonate / total CO 2 in the range of 18-24 mmol /l.

2. Anemia:

Anemia is a common problem in dogs and cats with chronic kidney disease. Observational studies have identified an association between anemia and survival in cats with CKD. While interventional studies have not been performed to determine the optimal treatment targets for hematocrit levels in dogs or cats, observations suggest that treating anemia associated with chronic kidney disease can improve the quality of life for pets. Treatment options for anemia due to CKD includes darbepoetin, HIF-PH inhibitors such as molidustat and blood transfusions.

Anemia can have multiple causes and animals should also be evaluated and treated for contributing factors, such as infections or excessive blood sampling.

I. Dogs with anemia due to CKD with a hematocrit of < 30% should be treated.

II. Dogs with persistent anemia due to CKD with a hematocrit between 30 and 35% should be treated.

3. Gastrointestinal Signs:

1. Treat vomiting and suspected nausea with anti-emetics such as maropitant or ondansetron.

2. Treat decreased appetite/weight and/or muscle loss with appetite stimulants such as capromorelin or mirtazapine.

3. Consider intermittent omeprazole in dogs with suspected gastric bleeding (eg melena, iron deficiency) or vomiting-induced esophagitis.

4. If pharmacological management of appetite is ineffective and/or supplemental hydration is required long-term, enteral feeding tube should be considered.

5. Give appropriate maintenance fluids parenterally as necessary to maintain hydration (see Footnote).

6. Drugs that rely predominantly on renal function for their clearance from the body should be used with caution in patients in Stage 3 CKD. It may be necessary to adjust the dose of these drugs (depending on their therapeutic indices) to avoid accumulation.

Where there is a discrepancy between creatinine and SDMA:

If serum or plasma SDMA is >54 µg/dl in a dog with a blood creatinine concentration of between 2.8 and 5 mg/dl (IRIS CKD Stage 3 based on creatinine), the patient should be staged an IRIS CKD Stage 4 patient and receive the recommended treatments for Stage 4 patients.

Stage 4 Canine patients:

Most dogs with Stage 4 CKD exhibit many extra-renal signs. Although it is still advisable to try to slow progression of CKD, treatments to improve quality of life become more important, especially management of dehydration, acidosis, vomiting, nausea and inappetance, weight loss and anemia. Treatment recommendations therefore include all of the above listed for Stages 1, 2 and 3, (listed again here for convenience) plus any additional steps indicated below.

1. Discontinue all potentially nephrotoxic drugs if possible.

2. Identify and treat any pre-renal or post-renal abnormalities.

3. Rule out any treatable conditions like pyelonephritis (any urinary tract infection should be regarded as a potential pyelonephritis and treated appropriately) and ureteral obstruction with radiographs and/or ultrasonography.

4. Measure blood pressure and urine protein to creatinine ratio (UP/C).

5. Clinical kidney diet therapy.

Management of dehydration:

These patients have decreased urine concentrating ability and therefore

1. Correct clinical dehydration/hypovolemia with isotonic, polyionic replacement fluid solutions (e.g., lactated Ringer's) IV or SQ, promptly as needed.

2. Have fresh water available at all times for drinking.

3. In addition, some of these dogs may require maintenance fluids administered routinely to maintain hydration (see below)

Systemic hypertension:

The blood pressure above which progressive kidney injury may be induced is unknown. Our goal is to reduce systolic blood pressure to <160 mm Hg and minimize the risk of extra-renal target organ damage (CNS, retinal, cardiac problems/damage). If there is no evidence of this but systolic blood pressure persistently exceeds 160 mm Hg, increasing the risk of such damage, treatment should be instituted.

1. Persistence of increased systolic blood pressure should be judged on multiple measurements made over the following time-scales in these blood pressure substages:

2. Hypertensive (moderate risk of future target organ damage) – systolic blood pressure 160 to 179 mm Hg, persistence demonstrated over 2 to 4 weeks.

3. Severely hypertensive (high risk of future target organ damage) – systolic blood pressure >180 mm Hg, persistence demonstrated over 1 to 2 weeks.

4. If evidence of target organ damage exists, dogs should be treated without the need to demonstrate persistently increased systolic blood pressure. Reducing blood pressure is a long term aim when managing the patient with CKD and a gradual and sustained reduction should be the goal, avoiding any sudden or severe decreases leading to hypotension.

It is recognized that some breeds (such as sight hounds) tend to have higher blood pressure (see Appendix 1) and that this may influence interpretation.

1. Dietary sodium (Na) reduction – there is no evidence that lowering dietary Na will reduce blood pressure. If dietary Na reduction is attempted, it should be accomplished gradually and in combination with pharmacological therapy.

2. Angiotensin converting enzyme inhibitor (ACEI, such as benazepril) therapy at standard dose rate.

3. Double the dose of ACEI (in some patients, increasing the dose may improve the antihypertensive effect).

4 .Combine ACEI and calcium channel blocker (CCB, such as amlodipine) treatment, especially if severely hypertensive.

5. Combine ACEI and CCB with an angiotensin receptor blocker (ARB, such as telmisartan) and/or hydralazine if additional treatment is required.

Note: Take care not to introduce ACEI/CCB with or without ARB treatment to unstable dehydrated dogs as glomerular filtration rate may drop precipitously if these drugs are introduced before the patient is adequately hydrated. The risk benefit analysis of combining ACEI with ARBs needs to be made on an individual dog basis and careful monitoring is required to ensure any deterioration in kidney function that is detected. The risk is higher for CKD stages 3 and 4 patients.

Monitoring response to antihypertensive treatment:

1. Hypertensive dogs normally require lifelong therapy and may require treatment adjustments. Serial monitoring is essential. After stabilization, monitoring should occur at least every 3 months.

2. Systolic blood pressure <120 mm Hg and/or clinical signs such as weakness or tachycardia indicate hypotension, which is to be avoided.

3. Blood creatinine concentration – reducing blood pressure may lead to small and persistent increases in creatinine (<45 μmol/l or 0.5 mg/dl increase) and/or SDMA (< 2 μg/dl), but a marked increase suggests an adverse drug effect.Progressively increasing creatinine concentrations indicate progressive kidney damage/disease.

Proteinuria:

Dogs in Stage 4 with urine protein to creatinine ratio (UP/C) >0.5 should be investigated for the disease processes leading to proteinuria. Those with confirmed and persistent renal proteinuria should be treated with anti-proteinuric measures. The goal of treatment is nuanced. Treatment should be aimed to have the reduction in proteinuria to the lowest UPC possible without doing harm.

Those with borderline proteinuria (0.2 to 0.5) require close monitoring.

1. Look for any concurrent associated disease process that may be treated/corrected.

2. Consider kidney biopsy (for dogs in stages 1 to 3, not stage 4) as a means of identifying underlying disease (see Appendix 2 and/or consult experts if unsure of indications for kidney biopsy).

3. Administer an angiotensin receptor blocker (ARB) and feed a clinical kidney diet.

4. Combination of an ACEI and diet with ARB if proteinuria is not controlled, should be done judiciously and cautiously, ideally under consultation with a veterinary nephrologist (see note 1 below).

5. Consider clopidogrel (1.1-3 mg/kg orally every 24 hours). If clopidogrel is not available, acetylsalicylic acid (2 to 5 mg/kg once daily) is an acceptable alternative (see note 2 below).

6. Monitor response to treatment / progression of disease:

- Stable blood creatinine concentration, decreasing UP/C and/or increasing serum albumin (if previously low)= good response.

  A UPC of < 0.5 is not achievable for many dogs with primary glomerular disease. For these dogs the goal should be a 50% reduction in UPC from baseline.

- Serially increasing blood creatinine concentrations and/or increasing UP/C = disease is progressing.

Ordinarily therapy will be maintained lifelong unless the underlying disease has been resolved in which case dose reduction whilst monitoring UP/C might be considered.

Note:

a. ACEI or ARB use is contraindicated in any animal that is clinically dehydrated and/or is showing signs of hypovolemia. Correct dehydration before using these drugs otherwise glomerular filtration rate may drop precipitously. The risk benefit analysis of combining ACEI with ARBs needs to be made on an individual dog basis and careful monitoring is required to ensure any deterioration in kidney function is detected. The risk is higher in CKD stages 3 and 4. Further detailed recommendations for diagnosis and management of glomerular disease in dogs can be found in the IRIS Consensus Statements published in Journal of Veterinary Internal Medicine in 2013 (supplement to volume 27). https://onlinelibrary.wiley.com/toc/19391676/2013/27/s1). ÿ

b. Dogs with PLN are at risk of thrombosis however it is not possible to predict the risk for thrombosis in the individual patient as serum albumin, antithrombin and degree of proteinuria are poorly associated with thrombotic risk. Tests such as thromboelastography, thrombin generation and markers of activated clotting can be used to document hypercoagulability, however, identification of a hypercoagulable state has not been shown to correlate to risk of developing thrombosis. Prothrombin and partiel thrombloplastin times (PT, PTT) also cannot be used to predict thrombotic risk.

   Antithrombotic therapy is indicated in dogs with PLN (CURATIVE Guidelines DOI: 10.1111/vec.12801). Clopidogrel (1-4mg/kg orally once daily) administration may be more effective than low dose acetylsalicylic acid (1-5mg/kg orally once daily) for thromboprophylaxis

Reduction of phosphate intake:

Evidence suggests that chronic reduction of phosphate intake to maintain a plasma phosphate concentration below 1.5 mmol/l (but not less than 0.9 mmol/l; <4.6 mg/dl but >2.7 mg/dl) is beneficial to patients with CKD.

A more realistic post-treatment target for dogs at Stage 4 is <1.9 mmol/l (6.0 mg/dl). The following measures can be introduced sequentially in an attempt to achieve this:

1. Dietary phosphate restriction (i.e. clinical kidney diet therapy).

2. If plasma phosphate concentration remains above 1.9 mmol/l (6.0 mg/dl) after dietary restriction, give enteric phosphate binders (such as aluminum hydroxide, aluminum carbonate, calcium carbonate, calcium acetate, lanthanum carbonate, ferric citrate) to effect, starting at 30-60 mg/kg/day in divided doses to be mixed with each meal (mixed with the food). The dose required will vary according to the amount of phosphate being fed and the stage of kidney disease. Treatment with phosphate binders should be to effect (as outlined above), with signs of toxicity limiting the upper dose rate possible in a given patient. Monitor serum calcium and phosphate concentrations every 4-6 weeks until stable and then every 12 weeks. Aluminum toxicity has been reported in a small number of dogs with stage 4 CKD. Clinical signs included muscle weakness, encephalopathy and microcytosis. If toxicity is suspected, serum aluminum levels can be measured and, if required, alternative phosphate binders are recommended. Hypercalcemia should be avoided-combinations of aluminum and calcium containing phosphate binders may necessary in some cases.

Further treatments aimed at improving quality of life (applicable to Stages 3 & 4)

1. If metabolic acidosis exists (blood bicarbonate or total CO2 <18 mmol/l) once the patient is stabilized on the clinical renal diet of choice, supplement with oral sodium bicarbonate (or potassium citrate if hypokalemic) to effect to maintain blood bicarbonate / total CO in the range of 18-24 mmol/l.

2. Anemia is a common problem in dogs and cats with chronic kidney disease. Observational studies have identified an association between anemia and survival in cats with CKD. While interventional studies have not been performed to determine the optimal treatment targets for hematocrit levels in dogs or cats, observations suggest that treating anemia associated with chronic kidney disease can improve the quality of life for pets. Treatment options for anemia due to CKD includes darbepoetin, HIF-PH inhibitors such as molidustat and blood transfusions.

Anemia can have multiple causes and animals should also be evaluated and treated for contributing factors, such as infections or excessive blood sampling.

I. Dogs with anemia due to CKD with a hematocrit of < 30% should be treated.

ii. Dogs with persistent anemia due to CKD with a hematocrit between 30 and 35% should be treated.

3. Gastrointestinal signs:

   I. Treat vomiting and suspected nausea with anti-emetics such as maropitant, or ondansetron.

   II. Treat decreased appetite/weight and/or muscle loss with appetite stimulants such as capromorelin or mirtazapine.

   III. Consider intermittent omeprazole in dogs with suspected gastric bleeding (e.g. melena, iron deficiency vomiting-induced esophagitis.

   IV. If pharmacological management of appetite is ineffective and/or supplemental hydration is required long-term, enteral feeding tube should be considered.

4. Give fluids parenterally as necessary to maintain hydration (see below).

5. Drugs that rely predominantly on kidney function for their clearance from the body should be used with caution in patients in Stage 4 CKD. It may be necessary to adjust the dose of these drugs (depending on their therapeutic indices) to avoid accumulation.

Further recommendations for Stage 4 patients:

1. Intensify efforts to prevent protein / calorie malnutrition. Consider feeding tube intervention (such as percutaneous esophagostomy or gastrostomy tube).

2. Intensify efforts to prevent dehydration. Feeding tubes can be used to administer fluids as well as food

3. Consider dialysis and/or renal transplantation.

Appendix 1

Adapted Blood Pressure Substaging

For most dogs, the IRIS blood pressure substages are as follows:

– Severely hypertensive (high risk of future target organ damage) – systolic blood pressure > 180 mm Hg
– Hypertensive (moderate risk of future target organ damage) – systolic blood pressure 160 to 179 mm Hg
– Prehypertensive (low risk of future target organ damage) – systolic blood pressure 140 to 159 mm Hg
– Normotensive (minimal risk of future target organ damage) – systolic blood pressure ‹140 mm Hg

It is recognized that some breeds, particularly sight hounds, tend to have higher blood pressure than many other breeds.
When dealing with these “high-pressure” breeds the classification of risk for future target organ damage might be adjusted as follows:

– High risk > 40 mm Hg above breed-specific reference range
– Moderate risk 20-40 mm Hg above breed-specific reference range
– Low risk 10-20 mm Hg above breed-specific reference range

Appendix 2

Reasons for Undertaking Renal Biopsy

1. Renomegaly
2. CKD in a young patient
3. Persistent and severe proteinuria (UP/C>2.0) in a non-azotemic patient
4. Worsening proteinuria in a CKD patient
5. Acute kidney injury, where renal biopsy may provide a prognostic indicator

Footnote

Maintenance fluids to maintain hydration status are low in sodium (30-40 mmol/l) and ideally have added potassium (about 13 mmol/l) to ensure daily requirements for fluid and electrolytes are met (e.g. Normosol-M® or 5% Dextrose plus 0.18% NaCl with added KCl).

Disclaimer

Although every effort has been made to ensure the completeness and accuracy of the information provided herein, the IRIS Board assumes no responsibility for the completeness or accuracy of the information.
All information is provided “as is” without any warranties, either expressed or implied.

All treatments for chronic kidney disease (CKD) need to be tailored to the individual patient. The following recommendations are useful starting points for most cats at each stage. Serial monitoring of these patients is ideal, and treatment should be adapted according to the response to treatment.
Note that staging of disease is undertaken **following diagnosis of CKD** – an increased blood creatinine or SDMA concentration alone is not diagnostic of CKD.

Treatment recommendations fall into two broad categories, namely:

1. Those that slow progression of CKD and so preserve remaining kidney function for longer

2. Those that address the quality of life of the cat, addressing the clinical signs of CKD

In general, at the early stages of CKD (stages 1 and 2), there are few clinical extra-renal signs of the disease and the therapeutic emphasis is on slowing progression. From stage 3 onwards, extra-renal signs become more common and more severe. The importance of administering treatments which address the clinical signs of CKD and improve the cat's quality of life assume greater importance and exceed the importance of treatments designed to slow progression by stage 4.

Some of the treatment recommendations are not authorized for use in all geographical regions and some may not be authorized for use in cats. Such recommended dose rates are therefore empirical. It is the treating veterinarian's duty to make a risk:benefit assessment for each patient prior to administering any treatment.

Treatment recommendations for Cats with Chronic Kidney Disease

Stage 1 Feline patients:

1. Discontinue all potentially nephrotoxic drugs if possible.

2. Identify and treat any pre-renal or post-renal abnormalities.

3. Rule out any treatable conditions like pyelonephritis and ureteral obstruction with radiographs and / or ultrasound.

4. Measure blood pressure and urine protein to creatinine ratio (UP/C).

Management of dehydration:

In these patients' urine concentrating ability may be somewhat impaired and therefore ensure:

* They always have fresh water available for drinking

* If become ill for any reason that leads to fluid losses, correct clinical dehydration with isotonic polyioinic replacement fluid solutions (e.g. lactated Ringer's) IV or SQ, promptly as needed

Systemic hypertension:

The blood pressure above which progressive renal injury may be induced is unknown. Our goal is to reduce systolic blood pressure to <160 mm Hg and to minimize the risk of extra-renal target organ damage (CNS, retinal, cardiac problems/damage). If there is no evidence of such damage but systolic blood pressure persistently exceeds 160 mm Hg, increasing the risk of this occurring, treatment should be instituted.

'Persistence' of increased systolic blood pressure should be judged on multiple measurements made at different clinic visits (over 1 to 2 weeks). The higher the systolic blood pressure the shorter the time frame over which persistence should be judged.

If evidence of target organ damage exists, cats should be treated without the need to demonstrate persistently increased systolic blood pressure. Reducing blood pressure is a long term aim in managing the patient with CKD and a gradual and sustained reduction should be the goal, avoiding any sudden or severe decreases leading to hypotension.

A logical stepwise approach to managing hypertension is as follows:

1. Dietary sodium (Na) reduction - there is no evidence that lowering dietary Na will reduce blood pressure. If dietary Na reduction is attempted, it should be accomplished gradually and in combination with pharmacological therapy.

2. Calcium channel blocker (CCB), such as amlodipine (0.125 to 0.25 mg/kg once daily) or angiotensin receptor blocker (ARB), telmisartan (2 mg/kg once daily).

3. Double the dose of amlodipine (0.25 to 0.5 mg/kg once daily) if this is treatment selected. Note telmisartan label does not advise a dose increase from 2 mg/kg once daily.

4. Combine amlodipine and telmisartan if either drug alone does not lead to adequate control of blood pressure.

Note: Take care not to introduce CCB/RAAS inhibitor treatment to unstable dehydrated cats as glomerular filtration rate may drop precipitously if these drugs are introduced before the patient is adequately hydrated.

Monitoring response to antihypertensive treatment:

Hypertensive cats normally require lifelong therapy and may require treatment adjustments. Serial monitoring is essential. After stabilization, monitoring should occur at least every 3 months.

Systolic blood pressure <120 mm Hg and/or clinical signs such as weakness or tachycardia indicate hypotension, which is to be avoided.

Blood creatinine concentration – reducing blood pressure may lead to small and persistent increases in creatinine concentration (<45 µmol/l or 0.5 mg/dl increase) or SDMA concentration (<2.0 µg/dl), but a marked increase suggests an adverse drug effect. Progressively increasing concentrations indicate progressive kidney damage/disease.

Proteinuria:

Cats in Stage 1 with UP/C >0.4 should be investigated for disease processes leading to proteinuria and treated with anti-proteinuric measures. Those with borderline proteinuria (UP/C 0.2 to 0.4) require close monitoring.

1. Look for any concurrent associated disease process that could be treated/corrected

2. Consider kidney biopsy as a means of identifying underlying disease (see Appendix and/or consult experts if unsure of indications for kidney biopsy)

3. Administer an RAAS inhibitor (angiotensin converting enzyme inhibitor (ACEI) or ARB) and feed a clinical renal diet.

4. Monitor response to treatment / progression of disease

    i. stable blood creatinine concentration and decreasing UP/C = good response

    ii. serially increasing blood creatinine concentration and /or increasing UP/C= disease is progressive

Ordinarily therapy will be maintained lifelong unless the underlying disease has been resolved in which case dose reduction whilst monitoring UP/C might be considered.

Note:

a. The use of a RAAS inhibitor is contraindicated in any cat that is clinically dehydrated or is showing signs of hypovolemia. Correct dehydration before using these drugs otherwise glomerular filtration rate may drop precipitously.

b. Cats with proteinuria and hypoalbuminemia likely share the same thromboembolic risk as dogs (although evidence is lacking in the published literature), but aspirin is difficult to use in cats to achieve a selective antiplatelet effect. If thromboembolic therapy is deemed necessary, clopidogrel (10 to 18.75 mg/day) is the drug of choice. Aspirin (1 mg/kg every 72 h) is an alternative (but see above).

c. Cats with serum phosphate within the IRIS target (4.5 mg/dl or 1.5 mmol/l) may be at increased risk of developing hypercalcemia when renal diets are introduced. Measurement of FGF23 may help to identify cats which would benefit from dietary phosphate restriction where plasma phosphate is in the target range. FGF23 >400 pg/ml in the absence of hypercalcemia, anemia or marked inflammatory disease is an indication to start dietary phosphate restriction. Monitor serum calcium and if total calcium exceeds 12 mg/dl (3 mmol/l) switch the cat to a less phosphate restricted diet.

d. If borderline proteinuria persists, antiproteinuric treatment could be offered, because the association between progression of CKD and proteinuria includes the borderline category. However, there is at present no evidence that intervention with anti-proteinuric drugs slows progression.

Stage 2 Feline patients:

All of the above listed for Stage 1 (listed here again for convenience) plus any additional steps indicated.

1. Discontinue all potentially nephrotoxic drugs if possible.

2. Identify and treat any pre-renal or post-renal abnormalities.

3. Rule out any treatable conditions like pyelonephritis and ureteral obstruction with radiographs and/or ultrasonography.

4. Measure blood pressure and urine protein to creatinine ratio (UP/C).

5. Consider feeding a clinical renal diet: this may be accomplished more easily early in the course of CKD, before inappetence develops.

Management of dehydration:

These patients often have decreased urine concentrating ability and therefore ensure:

- They always have fresh water available for drinking

- If cats become ill for any reason that leads to fluid losses, correct clinical dehydration with isotonic polyioinic replacement fluid solutions (e.g. lactated Ringer's) IV or SQ, promptly as needed

Systemic hypertension:

The blood pressure above which progressive renal injury may be induced is unknown. Our goal is to reduce systolic blood pressure to <160 mm Hg and to minimize the risk of extra-renal target organ damage (CNS, retinal, cardiac problems/damage). If there is no evidence of this but systolic blood pressure persistently exceeds 160 mm Hg, increasing the risk of such damage, treatment should be instituted.

'Persistence' of increased systolic blood pressure should be judged on multiple measurements made at different clinic visits (over 1 to 2 weeks). The higher the systolic blood pressure the shorter the time frame over which persistence should be judged.

If evidence of target organ damage exists, cats should be treated without the need to demonstrate persistently increased systolic blood pressure. Reducing blood pressure is a long term aim in managing the patient with CKD and a gradual and sustained reduction should be the goal, avoiding any sudden or severe decreases leading to hypotension.

A logical stepwise approach to managing hypertension is as follows:

1. Dietary sodium (Na) reduction - there is no evidence that lowering dietary Na will reduce blood pressure. If dietary Na reduction is attempted, it should be accomplished gradually and in combination with pharmacological therapy.

2. Calcium channel blocker (CCB) such as amlodipine (0.125 to 0.25 mg/kg once daily) or angiotensin receptor blocker (ARB), telmisartan (2 mg/kg once daily).

3. Double the dose of amlodipine (0.25 to 0.5 mg/kg once daily) if this is treatment selected. Note telmisartan label does not advise a dose increase from 2 mg/kg once daily.

4. Combine amlodipine and telmisartan if either drug on its own does not lead to adequate control of blood pressure

Note: Take care not to introduce CCB/RAAS inhibitor treatment to unstable dehydrated cats as glomerular filtration rate may drop precipitously if these drugs are introduced before the patient is adequately hydrated.

Monitoring response to antihypertensive treatment:

Hypertensive cats normally require lifelong therapy and may require treatment adjustments. Serial monitoring is essential. After stabilization, monitoring should occur at least every 3 months.

Systolic blood pressure <120 mm Hg and/or clinical signs such as weakness or tachycardia indicate hypotension, which is to be avoided.

Blood creatinine concentration – reducing blood pressure may lead to small and persistent increases in creatinine (< 45 μmol/l or 0.5 mg/dl increase) or SDMA concentration (< 2.0 μg/dl), but a marked increase suggests an adverse drug effect. Progressively increasing creatinine concentrations indicate progressive kidney damage/disease.

Proteinuria:

Cats in Stage 2 with UP/C >0.4 should be investigated for disease processes leading to proteinuria and treated with anti-proteinuric measures. Those with borderline proteinuria require close monitoring.

1. Look for any concurrent associated disease process that may be treated/corrected.

2. Consider kidney biopsy as a means of identifying underlying disease (see Appendix and/or consult experts if unsure of indications for kidney biopsy).

3. Administer an RAAS inhibitor (ACEI or ARB) and feed a clinical renal diet.

4. Monitor response to treatment/progression of disease:

   - stable blood creatinine concentration and decreasing UP/C = good response.

   - serially increasing creatinine concentrations and/or increasing UP/C = disease is progressing.

Ordinarily therapy will be maintained lifelong unless the underlying disease has been resolved in which case, dose reduction while monitoring UP/C might be considered.

Note:

a. Use of an RAAS inhibitor is contraindicated in any animal that is clinically dehydrated and/or showing signs of hypovolemia. Correct dehydration before using these drugs otherwise glomerular filtration rate may drop precipitously.

b. Cats with proteinuria and hypoalbuminemia likely share the same thromboembolic risk as dogs, (although evidence is lacking in the published literature), but aspirin is difficult to use in cats to achieve a selective antiplatelet effect. If thromboembolic therapy is deemed necessary, clopidogrel (10 to 18.75 mg/day) is the drug of choice. Aspirin (1 mg/kg every 72 h) is an alternative (but see above).

c. Cats with serum phosphate within the IRIS target may be at increased risk of developing hypercalcemia when renal diets are introduced. Measurement of FGF23 may help to identify cats which would benefit from dietary phosphate restriction where plasma phosphate is in the target range. FGF23 > 400 pg/ml in the absence of hypercalcaemia, anemia or marked inflammatory disease is an indication to start dietary phosphate restriction. Monitor serum calcium and if total calcium exceeds 12mg/dl (3mmol/l) switch the cat to a less phosphate restricted diet.

d. If borderline proteinuria is persistent, antiproteinuric treatment could be offered. The rationale for doing so is that there is an association between progression of CKD and proteinuria which extends into the borderline category. There is no evidence that intervention with anti-proteinuric drugs slows progression, however.

Reduction of phosphate intake:

Many cats in Stage 2 will have normal plasma phosphate concentrations but will have increased plasma FGF23 concentration. Evidence suggests that chronic reduction of phosphate intake to maintain a plasma phosphate concentration below 1.5 mmol/l or 4.5 mg/dl (but not less than 0.9 mmol/l or 2.7 mg/dl) is beneficial to patients with CKD.

Measurement of FGF23 may help to identify cats which would benefit from dietary phosphate restriction if plasma phosphate is within the IRIS target range. FGF23 >400 pg/ml in the absence of hypercalcaemia, anemia or marked inflammatory disease is an indication to start dietary phosphate restriction.

The following measures can be introduced sequentially in an attempt to achieve this:

1. Dietary phosphate restriction (i.e., clinical renal diet therapy).

2. If plasma phosphate concentration remains above 1.5 mmol/l (4.5 mg/dl) after dietary restriction, give enteric phosphate binders (such as aluminum hydroxide, aluminum carbonate, calcium carbonate, calcium acetate, lanthanum carbonate) to effect, starting at 30-60 mg/kg/day in divided doses to be mixed with each meal (mixed with the food). The dose required will vary according to the amount of phosphate being fed and the stage of kidney disease. Treatment with phosphate binders should be to effect (as outlined above), with signs of toxicity limiting the upper dose rate possible in a given patient. Monitor serum calcium and phosphate concentrations every 4-6 weeks until stable and then every 12 weeks. Once serum phosphate is within the target range for stage 2 (1.5 mmol/l or 4.5 mg/dl) measurement of serum FGF23 may assist in determining whether further phosphate restriction would be beneficial to the cat.

   If FGF23 is >700 pg/ml, further phosphate restriction should be applied (e.g. by increasing the dose of phosphate binder) provided there is no evidence of hypercalcaemia, marked anemia or severe inflammatory disease (all of which can increase FGF23 independently of mineral bone disturbance). Excellent control of mineral bone disturbance would be indicated by serum FGF23 of <500 pg/ml.

   Microcytosis and/or generalized muscle weakness and neurological signs suggest aluminum toxicity if using an aluminum containing binder – switch to another form of phosphate binder should this occur. It should be noted, however, unlike the dog, aluminum toxicity has not been reported in the cat and it is possible to measure blood aluminum levels to confirm suspected cases. Hypercalcemia should be avoided – combinations of aluminum and calcium containing phosphate binders may be necessary in some cases.

Additional recommendations for stage 2 patients

1. If the patient is hypokalemic, then potassium citrate or potassium gluconate should be supplemented to effect (typically 1-2 mmol/kg/day).

2. Treat vomiting / decreased appetite / nausea / weight and/or muscle loss with an antiemetic /appetite stimulant / anti-nausea agent (such as maropitant, ondansetron, oral or transdermal mirtazapine, or capromorelin).

3. Anemia is a common problem in dogs and cats with chronic kidney disease. Observational studies have identified an association between anemia and survival in cats with CKD. While interventional studies have not been performed to determine the optimal treatment targets for hematocrit levels in dogs or cats, observations suggest that treating anemia associated with chronic kidney disease can improve the quality of life for pets. Treatment options for anemia due to CKD includes darbepoetin, HIF-PH inhibitors such as molidustat and blood transfusions. Anemia can have multiple causes and animals should also be evaluated and treated for contributing factors, such as infections or excessive blood sampling.

   a. Cats with anemia due to CKD with a hematocrit of < 25 % should be treated.

   b. Cats with persistent anemia due to CKD with a hematocrit of 25 to 28% should be treated.

   In a euhydrated cat with Stage 2 CKD, anemia of <25% is less likely to be attributable to CKD and the patient should be assessed for other disease processes. If further workup does not reveal any other underlying cause, then the anemia is most likely attributable to CKD and should be treated.

4. If there is marked muscle loss, measure serum SDMA and base staging and treatment recommendations on SDMA rather than creatinine as it is not affected by muscle wasting. In addition, evaluate for concurrent diseases leading to vomiting and weight loss.

5. Evidence suggests oral mirtazapine (1.88 mg/cat every 48 h for 3 weeks) reduces vomiting, improves appetite and leads to weight gain in cats showing signs of inappetence and vomiting in this stage. Maropitant (1 mg/kg daily for 2 weeks) reduced vomiting but did not lead to weight gain/ improved appetite. Further investigations are needed on the use of these and other drugs to determine whether they are useful for managing gastrointestinal disturbances in cats with CKD and uremia when administered longer term.

Where there is a discrepancy between creatinine and SDMA:

If serum or plasma SDMA is persistently > 25 ug/dl in a cat with serum creatinine between 1.6 and 2.8 mg/dl (IRIS CKD stage 2 based on creatinine), stage and treat the cat as an IRIS stage 3 CKD patient.

Stage 3 Feline patients:

The range of presentations for cats in Stage 3 is likely to be wide, from no clinical signs to quite marked extra-renal clinical signs. The main treatments mentioned so far for stages 1 and 2 are aimed at slowing progression of CKD also apply in stage 3 and may be the only therapies needed for cats with no or mild extra-renal signs. However, treatments designed to improve the quality of life of the cat become more important, the more extra-renal signs are present. These include treatments to address dehydration, nausea and vomiting, anemia and acidosis.

Treatments include all of the steps listed for Stage 1 and 2 (listed here again for convenience) plus any additional steps indicated.

1. Discontinue all potentially nephrotoxic drugs if possible.

2. Identify and treat any pre-renal or post-renal abnormalities.

3. Rule out any treatable conditions like pyelonephritis and ureteral obstruction with radiographs and/or ultrasonography.

4. Measure blood pressure and urine protein to creatinine ratio (UP/C).

5. Feed a clinical renal diet.

Management of dehydration:

These patients have decreased urine concentrating ability and therefore

- correct clinical dehydration/hypovolemia with isotonic, polyionic replacement fluid solutions (e.g., lactated Ringer's) IV or SQ as needed.

- have fresh water available at all times for drinking.

- In addition, some of these cats may require maintenance fluids administered routinely to maintain hydration (see below)

Systemic hypertension:

The blood pressure above which progressive renal injury may be induced is unknown.
Our goal is to reduce systolic blood pressure to <160 mm Hg and to minimize the risk of extra-renal target organ damage (CNS, retinal, cardiac problems/damage). If there is no evidence of this but systolic blood pressure persistently exceeds 160 mm Hg increasing the risk of extra-renal target organ damage, treatment should be instituted.

'Persistence' of increased systolic blood pressure should be judged on multiple measurements made at different clinic visits (over 1 to 2 weeks). The higher the systolic blood pressure the shorter the time frame over which persistence should be judged.

If evidence of target organ damage exists, cats should be treated without the need to demonstrate persistently increased systolic blood pressure. Reducing blood pressure is a long term aim in managing the patient with CKD and a gradual and sustained reduction should be the goal avoiding sudden or severe decreases leading to hypotension.

A logical stepwise approach to managing hypertension is as follows:

1. Dietary sodium (Na) reduction – there is no evidence that lowering dietary Na will reduce blood pressure. If dietary Na reduction is attempted, it should be accomplished gradually and in combination with pharmacological therapy.

2. Calcium channel blocker (CCB) such as amlodipine (0.125 to 0.25 mg/kg once daily) or angiotensin receptor blocker (ARB), telmisartan (2 mg/kg once daily)

3. Double the dose of amlodipine (0.25 to 0.5 mg/kg once daily) if this is treatment selected. Note telmisartan label does not advise a dose increase from 2 mg/kg once daily.

4. Combine amlodipine and telmisartan if either drug on its own does not lead to adequate control of blood pressure.

Note: Take care not to introduce CCB/RAAS inhibitor treatment to unstable dehydrated cats as glomerular filtration rate may drop precipitously if these drugs are introduced before the patient is adequately hydrated.

Monitoring response to antihypertensive treatment:

Hypertensive cats normally require lifelong therapy and may require treatment adjustments. Serial monitoring is essential. After stabilization, monitoring should occur at least every 3 months.

Systolic blood pressure <120 mm Hg and/or clinical signs such as weakness or tachycardia indicate hypotension, which is to be avoided.

Blood creatinine concentration – reducing blood pressure may lead to small and persistent increases in creatinine concentration (<45 µmol/l or 0.5 mg/dl increase) or SDMA concentration (<2.0 µg/dl), but a marked increase suggests an adverse drug effect. Progressively increasing concentrations indicate progressive kidney damage/disease.

Proteinuria:

Cats in Stage 3 with urine protein to creatinine ratio (UP/C) >0.4 should be investigated for disease processes leading to proteinuria and treated with anti-proteinuric measures. Those with borderline proteinuria (0.2 to 0.4) require close monitoring.

1. Look for any concurrent associated disease process that may be treated/corrected.

2. Consider kidney biopsy as a means of identifying underlying disease (see Appendix and/or consult experts if unsure of indications for kidney biopsy).

3. Administer an RAAS inhibitor (ACEI or ARB) and feed a clinical renal diet.

4. Monitor response to treatment / progression of disease:

    - stable blood creatinine concentration and decreasing UP/C = good response.

    - serially increasing creatinine concentrations and/or increasing UP/C = disease is progressing.

    Ordinarily therapy will be maintained life-long unless the underlying disease has been resolved in which case dose reduction whilst monitoring UP/C might be considered.

Note:

a. Use of an RAAS inhibitor is contraindicated in any animal that is clinically dehydrated and/or is showing signs of hypovolemia. Correct dehydration before using these drugs otherwise glomerular filtration rate may drop precipitously.

b. Cats with proteinuria and hypoalbuminemia likely share the same thromboembolic risk as dogs, (although evidence is lacking in the published literature), but aspirin is difficult to use in cats to achieve a selective antiplatelet effect. If thromboembolic therapy is deemed necessary, clopidogrel (10 to 18.75 mg/day) is the drug of choice. Aspirin (1 mg/kg every 72 h) is an alternative (but see above).

c. Cats with serum phosphate within the IRIS target may be at increased risk of developing hypercalcemia when renal diets are introduced. Monitor serum calcium and if total calcium exceeds 12mg/dl (3mmol/l) switch to a less phosphate restricted diet.

d. If borderline proteinuria is persistent, antiproteinuric treatment could be offered. The rationale for doing so is that there is an association between progression of CKD and proteinuria which extends into the borderline category. However, there is at present no evidence that intervention with anti-proteinuric drugs slows progression.

Reduction of phosphate intake:

Evidence suggests that chronic reduction of phosphate intake to maintain a plasma phosphate concentration below 1.5 mmol/l (4.5 mg/dl) but not less than 0.9 mmol/l (2.7 mg/dl) is beneficial to patients with CKD.

A more realistic post-treatment target plasma phosphate concentration for cats at Stage 3 is <1.6 mmol/l (5.0 mg/dl).

The following measures can be introduced sequentially in an attempt to achieve this:

1. Dietary phosphate restriction (i.e., clinical renal diet therapy).

2. If plasma phosphate concentration remains above 1.6 mmol/l (5 mg/dl) after dietary estriction, give enteric phosphate binders (such as aluminum hydroxide, aluminum carbonate, calcium carbonate, calcium acetate, lanthanum carbonate) to effect, starting at 30-60 mg/kg/day in divided doses to be mixed with each meal (mixed with the food). The dose required will vary according to the amount of phosphate being fed and the IRIS stage. Treatment with phosphate binders should be to effect (as outlined above), with signs of toxicity limiting the upper dose rate possible in a given patient. Monitor serum calcium and phosphate concentrations every 4-6 weeks until stable and then every 12 weeks.

   Once serum phosphate is in the target range for stage 3 (<1.6 mmol/l or 5 mg/dl) measurement of serum FGF23 may assist in determining whether further phosphate restriction would be (e.g. by increasing the dose of phosphate binder) provided there is no evidence of hypercalcemia, marked anemia or severe inflammatory disease (all of which can increase FGF23 independently of mineral bone disturbance).

   Microcytosis and/or generalized muscle weakness and neurological signs suggest aluminum toxicity if using an aluminum containing binder– switch to another form of phosphate binder should this occur. It should be noted, however, unlike the dog, aluminum toxicity has not been reported in the cat and it is possible to measure blood aluminum levels to confirm suspected cases. Hypercalcemia should be avoided – combinations of aluminum and calcium containing phosphate binders may be necessary in some cases.

Additional recommendations for Stage 3 patients:

1. If the patient is hypokalemic, then potassium gluconate or potassium citrate should be supplemented to effect (typically 1-2 mmol/kg/day).

2. Anemia is a common problem in dogs and cats with chronic kidney disease. Observational studies have identified an association between anemia and survival in cats with CKD. While interventional studies have not been performed to determine the optimal treatment targets for hematocrit levels in dogs or cats, observations suggest that treating anemia associated with chronic kidney disease can improve the quality of life for pets. Treatment options for anemia due to CKD includes darbepoetin, HIF-PH inhibitors such

as molidustat and blood transfusions. Anemia can have multiple causes and animals should also be evaluated and treated for contributing factors, such as infections or excessive blood sampling.

i. Cats with anemia due to CKD with a hematocrit of < 25 % should be treated.

ii. Cats with persistent anemia due to CKD with a hematocrit of 25 to 28% should be treated.

3. If metabolic acidosis exists (blood bicarbonate or total CO_2 <16 mmol/l) once the patient is stabilized on the diet of choice, supplement with oral sodium bicarbonate, (or potassium citrate if hypokalemic) to effect to maintain blood bicarbonate / total CO_2 in the range of 16-24 mmol/l.

4. Treat vomiting / decreased appetite / nausea / weight and/or muscle loss with an antiemetic / appetite stimulant / antinausea agent (such as maropitant, ondansetron or oral or transdermal mirtazapine or capromorelin). Evidence suggests mirtazapine (1.88 mg/cat every 48 h for 3 weeks) reduces vomiting, improves appetite and leads to weight gain in cats showing signs of inappetence and vomiting in this stage. Maropitant (1 mg/kg daily for 2 weeks) reduced vomiting but did not lead to weight gain/ improved appetite. Further investigations are needed on the use of these and other drugs to determine whether they are useful for managing gastrointestinal disturbances in cats with CKD and uremia when administered longer term. If pharmacological management of appetite is ineffective and/or supplemental hydration is required long-term, enteral feeding tube should be considered.

5. Give appropriate maintenance fluids parenterally as necessary to maintain hydration (see Footnote on page 22).

6. Drugs that rely predominantly on renal function for their clearance from the body should be used with caution in patients in Stage 3 CKD. It may be necessary to adjust the dose of these drugs (depending on their therapeutic indices) to avoid accumulation.

Where there is a discrepancy between creatinine and SDMA:

If serum or plasma SDMA is persistently >35 µg/dl in a cat with serum creatinine between 2.8 and 5 mg/dl (IRIS CKD stage 3 based on creatinine) stage and treat the cat as an IRIS CKD Stage 4 patient.

Stage 4 Feline patients:

Most cats with stage 4 CKD have many extra-renal signs present. Although it is still important to administer treatments which slow progression of CKD, the importance of improving quality of life for these cats is greater at this stage.

Therapies directly addressing clinical signs and aimed at improving quality of life include management of dehydration, acidosis, vomiting and nausea and anemia.

Treatments include all of the steps listed for listed for Stages 1, 2 and 3 (listed again for convenience) plus any additional steps indicated.

1. Discontinue all potentially nephrotoxic drugs if possible.
2. Identify and treat any pre-renal or post-renal abnormalities.
3. Rule out any treatable conditions like pyelonephritis and ureteral obstruction with radiographs and / or ultrasound.
4. Measure blood pressure and urine protein to creatinine ratio (UP/C).
5. Feed a clinical renal diet.

Management of dehydration:

These patients have decreased urine concentrating ability and therefore ensure:

- correct clinical dehydration/hypovolemia with isotonic, polyionic replacement fluid solutions (eg lactated Ringer's) IV or SQ, promptly as needed.
- have fresh water available at all times for drinking.
- In addition, these cats may require maintenance fluids administered routinely to maintain hydration (see Footnote on page 22).

Systemic hypertension:

The blood pressure above which progressive renal injury may be induced is unknown. Our goal is to reduce systolic blood pressure to <160 mm Hg and to minimize the risk of extra-renal target organ damage (CNS, retinal, cardiac problems/damage). If there is no evidence of this but systolic blood pressure persistently exceeds 160 mm Hg, increasing the risk of such damage, treatment should be instituted.

'Persistence' of increased systolic blood pressure should be judged on multiple measurements made at different clinic visits (over 1 to 2 weeks). The higher the systolic blood pressure the shorter the time frame over which persistence should be judged.

If evidence of target organ damage exists, cats should be treated without the need to demonstrate persistently increased systolic blood pressure. Reducing blood pressure is a long term aim in managing the patient with CKD and a gradual and sustained reduction should be the goal, avoiding any sudden or severe decreases leading to hypotension.

A logical stepwise approach to managing hypertension is as follows:

1. Dietary sodium (Na) reduction - there is no evidence that lowering dietary Na will reduce blood pressure. If dietary Na reduction is attempted, it should be accomplished gradually and in combination with pharmacological therapy.

2. Calcium channel blocker (CCB) such as amlodipine (0.125 to 0.25 mg/kg once daily) or angiotensin receptor blocker (ARB), telmisartan (2 mg/kg once daily).

3. Double the dose of amlodipine (0.25 to 0.5 mg/kg once daily) if this is treatment selected. Note telmisartan label does not advise a dose increase from 2 mg/kg once daily.

4. Combine amlodipine and telmisartan if either drug on its own does not lead to adequate control of blood pressure.

Note: Take care not to introduce CCB/RAAS inhibitor treatment to unstable dehydrated cats as glomerular filtration rate may drop precipitously if these drugs are introduced before the patient is adequately hydrated. The benefit of combining drugs to try to reduce blood pressure to below the target should be weighed against the high risk in a stage 4 cat of precipitating an azotemic crisis should be carefully considered.

Monitoring response to antihypertensive treatment:

Hypertensive cats normally require lifelong therapy and may require treatment adjustments. Serial monitoring is essential. After stabilization, monitoring should occur at least every 3 months.

Systolic blood pressure <120 mm Hg and/or clinical signs such as weakness or tachycardia indicate hypotension, which is to be avoided.

Blood creatinine concentration – reducing blood pressure may lead to small and persistent increases in creatinine (<45 μmol/l or 0.5 mg/dl increase) or SDMA concentration (<2.0 μg/dl), but a marked increase suggests an adverse drug effect. Progressively increasing concentrations indicate progressive kidney damage/disease.

Proteinuria:

Cats in Stage 4 with urine protein to creatinine ratio (UP/C) >0.4 should be investigated for disease processes leading to proteinuria and treated with anti-proteinuric measures.

Those with borderline proteinuria (0.2 to 0.4) require close monitoring.

1. Look for any concurrent associated disease process that may be treated/corrected.
2. Administer an RAAS inhibitor (ACEI or ARB) and feed a clinical renal diet.
3. Monitor response to treatment / progression of disease:
   - stable blood creatinine concentration and decreasing UP/C = good response.
   - serially increasing creatinine concentrations and/or increasing UP/C = disease is progressing.

Ordinarily therapy will be maintained lifelong unless the underlying disease has been resolved, in which case dose reduction whilst monitoring UP/C might be considered.

Note:

a. Use of an RAAS inhibitor is contraindicated in any animal that is clinically dehydrated and/or is showing signs of hypovolemia. Correct dehydration before using these drugs otherwise glomerular filtration rate may drop precipitously.

b. Cats with proteinuria and hypoalbuminemia likely share the same thromboembolic risk as dogs, (although evidence is lacking in the published literature), but aspirin is difficult to use in cats to achieve a selective antiplatelet effect. If thromboembolic therapy is deemed necessary, clopidogrel (10 to 18.75 mg/day) is the drug of choice. Aspirin (1 mg/kg every 72 h) is an alternative (but see above).

c. Cats with serum phosphate within the IRIS target may be at increased risk of developing hypercalcemia when renal diets are introduced. Monitor serum calcium and if total calcium exceeds 12mg/dl (3mmol/l) switch to a less phosphate restricted diet.

d. If borderline proteinuria is persistent, antiproteinuric treatment could be offered. The rationale for doing so is that there is an association between progression of CKD and proteinuria which extends into the borderline category. However, there is at present no evidence that intervention with anti-proteinuric drugs slows progression.

Reduction of phosphate intake:

Evidence suggests that chronic reduction of phosphate intake to maintain a plasma phosphate concentration below 1.5 mmol/l (4.5 mg/dl) but above 0.9 mmol/l (2.7 mg/dl) is beneficial to patients with CKD.

A more realistic post-treatment target plasma phosphate concentration for cats at Stage 4 is <1.9 mmol/l (6.0 mg/dl).

The following measures can be introduced sequentially in an attempt to achieve this:

1. Dietary phosphate restriction (i.e., clinical renal diet therapy).

2. If plasma phosphate concentration remains above 1.9 mmol/l (6.0 mg/dl) after dietary restriction, give enteric phosphate binders (such as aluminum hydroxide, aluminum carbonate, calcium carbonate, calcium acetate, lanthanum carbonate) to effect, starting at 30-60 mg/kg/day in divided doses to be mixed with each meal (mixed with the food). The dose required will vary according to the amount of phosphate being fed and the stage of CKD.

   Treatment with phosphate binders should be to effect (as outlined above), with signs of toxicity limiting the upper dose rate possible in a given patient. Monitor serum calcium and phosphate concentrations every 4-6 weeks until stable and then every 12 weeks

   Once serum phosphate is in the target range for stage 4 (<1.9 mmol/l or 6 mg/dl) measurement of serum FGF23 may assist in determining whether further phosphate restriction would be beneficial to the cat. If FGF23 is >700 pg/ml, further phosphate restriction should be applied (e.g. by increasing the dose of phosphate binder) provided there is no evidence of hypercalcemia, marked anemia or severe inflammatory disease (all of which can increase FGF23 independently of mineral bone disturbance).

   Microcytosis and/or generalized muscle weakness and neurological signs suggest aluminum toxicity if using an aluminum containing binder – switch to another form of phosphate binder should this occur. It should be noted, however, unlike the dog, aluminum toxicity has not been reported in the cat and it is possible to measure blood aluminum levels to confirm suspected cases. Hypercalcemia should be avoided – combinations of aluminum and calcium containing phosphate binders may be necessary in some cases.

Additional recommendations for stage 4 patients:

1. If the patient is hypokalemic, then potassium gluconate or potassium citrate should be supplemented to effect
   (typically 1-2 mmol/kg/day).

2. Anemia is a common problem in dogs and cats with chronic kidney disease. Observational studies have identified an association between anemia and survival in cats with CKD. While interventional studies have not been performed to determine the optimal treatment targets for hematocrit levels in dogs or cats, observations suggest that treating anemia associated with chronic kidney

disease can improve the quality of life for pets. Treatment options for anemia due to CKD includes darbepoetin, HIF-PH inhibitors such as molidustat and blood transfusions. Anemia can have multiple causes and animals should also be evaluated and treated for contributing factors, such as infections or excessive blood sampling.

a. Cats with anemia due to CKD with a hematocrit of < 25 % should be treated.

b. Cats with persistent anemia due to CKD with a hematocrit of 25 to 28% should be treated.

3. If metabolic acidosis exists (blood bicarbonate or total CO_2 <16 mmol/l) once the patient is stabilized on the diet of choice, supplement with oral sodium bicarbonate (or potassium citrate if hypokalemic) to effect to maintain blood bicarbonate / total CO_2 in the range of 16-24 mmol/l.

4. Treat vomiting / decreased appetite / nausea / weight and/or muscle loss with an antiemetic / appetite stimulant / antinausea agent (such as maropitant, ondansetron or ‘oral or transdermal mirtazapine or capromorelin). Evidence suggests oral mirtazapine (1.88 mg/cat every 48 h for 3 weeks) reduces vomiting, improves appetite and leads to weight gain in cats showing signs of inappetence and vomiting in this stage. Maropitant (1 mg/kg daily for 2 weeks) reduced vomiting but did not lead to weight gain/ improved appetite. Further investigations are needed on the use of these and other drugs to determine whether they are useful for managing gastrointestinal disturbances in cats with CKD and uremia when administered longer term. If pharmacological management of appetite is ineffective and/or supplemental hydration is required long-term, enteral feeding tube should be considered.

5. Give appropriate maintenance fluids parenterally as necessary to maintain hydration (see Footnote on page 22).

6. Intensify efforts to prevent protein / calorie malnutrition. Consider feeding tube intervention (e.g., percutaneous gastrostomy tube).

7. Intensify efforts to prevent dehydration. Feeding tubes can be used to administer fluids as well as food.

8. Consider omeprazole in cats in the uncommon situation where GI bleeding is suspected (e.g. melena, iron deficiency).

9. Consider dialysis and /or renal transplantation.

Drugs that rely predominantly on renal function for their clearance should be used with caution in Stage 4 CKD patients. It may be necessary to adjust the dose of these drugs (depending on their therapeutic indices) to avoid accumulation.

Appendix

Reasons for Undertaking Renal Biopsy

1. Renomegaly
2. CKD in a young patient
3. Persistent and severe proteinuria (UP/C>2.0) in a non-azotemic patient
4. Worsening proteinuria in a CKD patient
5. Acute kidney injury, where renal biopsy may provide a prognostic indicator

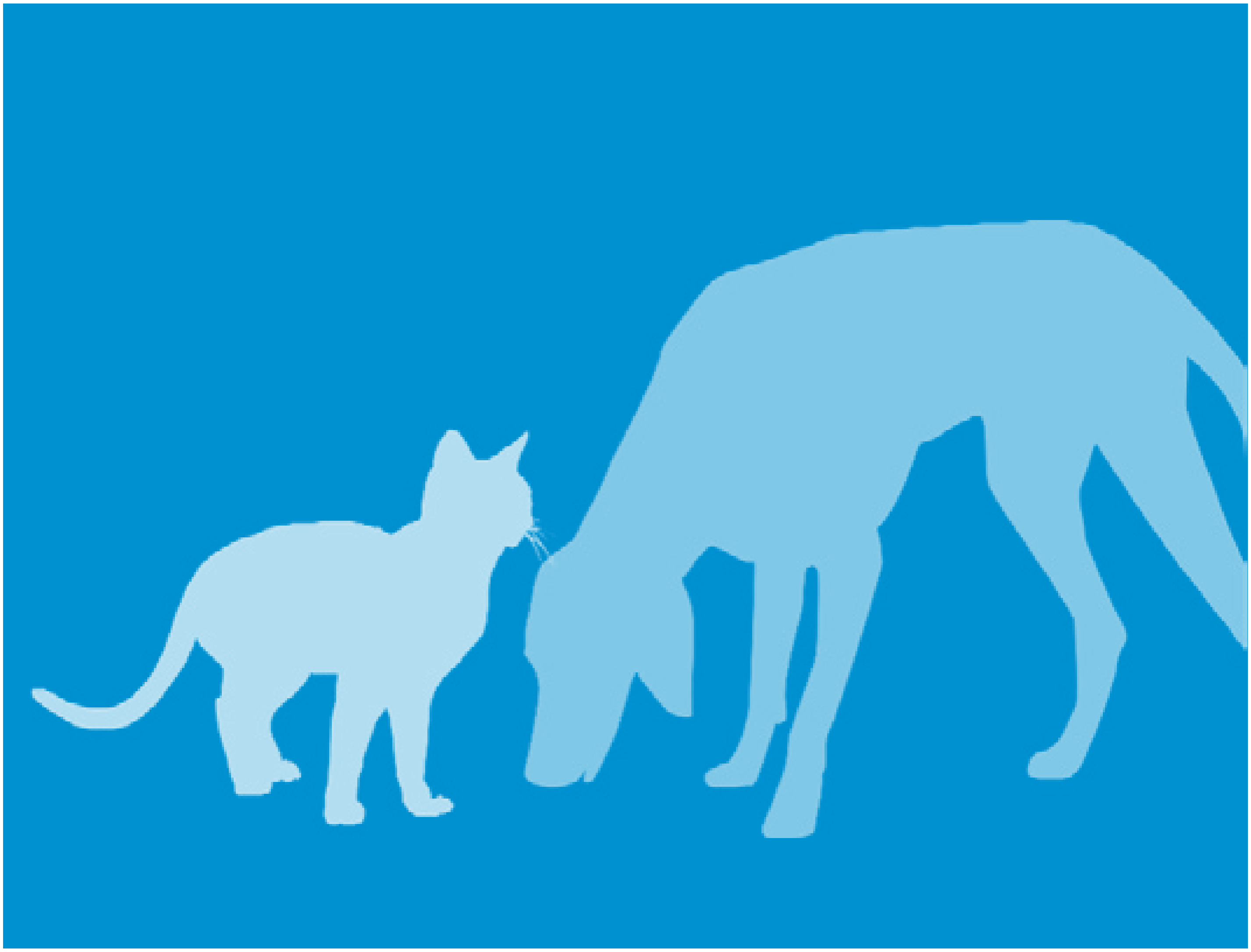
Footnote

Maintenance fluids to maintain hydration status are low in sodium (30-40 mmol/l) and ideally have added potassium (about 13 mmol/l) to ensure daily requirements for fluid and electrolytes are met (e.g. Normosol-M® or 5% Dextrose plus 0.18% NaCl with added KCl).

Disclaimer

Although every effort has been made to ensure the completeness and accuracy of the information provided herein, the IRIS Board assumes no responsibility for the completeness or accuracy of the information.
All information is provided “as is” without any warranties, either expressed or implied.

Chronic kidney disease (CKD) is diagnosed based on evaluation of all available clinical and diagnostic information in a stable patient. Following diagnosis of CKD, the IRIS Board recommends using serum creatinine or SDMA (ideally both) to stage CKD with substaging based on assessment of arterial blood pressure and proteinuria.



Step 1: Diagnose CKD

Clinical signs and physical examination findings worsen with increasing severity of kidney disease

Clinical presentation

Consider age, sex, breed predispositions, and relevant historical information, including medication history, toxin/toxicant exposure, and diet.
Can be subclinical in early stage CKD. Signs may include polyuria, polydipsia, weight loss, decreased appetite, lethargy, dehydration, vomiting, and bad breath.

Physical examination findings

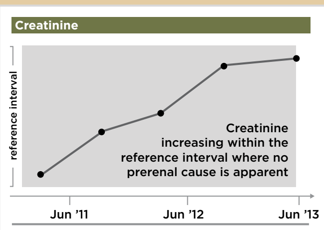
Can be normal in early stage CKD. Findings may include palpable kidney abnormalities, evidence of weight loss, dehydration, pale mucous membranes, uremic ulcers, evidence of hypertension, i.e., retinal hemorrhages/ detachment.

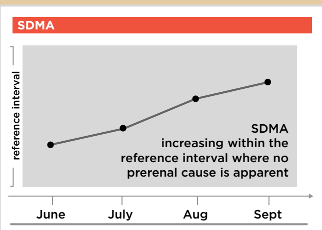
To diagnose Stage 1 and early Stage 2 CKD

OR

One or more of these diagnostic findings:

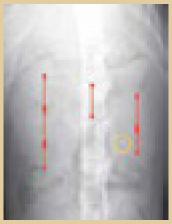
1



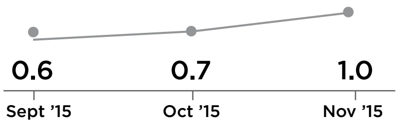


2 Persistent increased SDMA* >14 µg/dL

3 Abnormal kidney imaging



4 Persistent renal proteinuria
UPC >0.5 in dogs; UPC >0.4 in cats

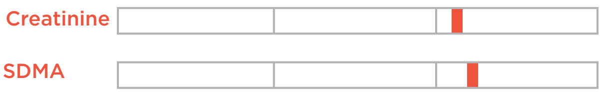


See www.iris-kidney.com for more detailed staging, therapeutic, and management guidelines.

To diagnose more advanced CKD (late Stage 2–4)

Both of these diagnostic findings:

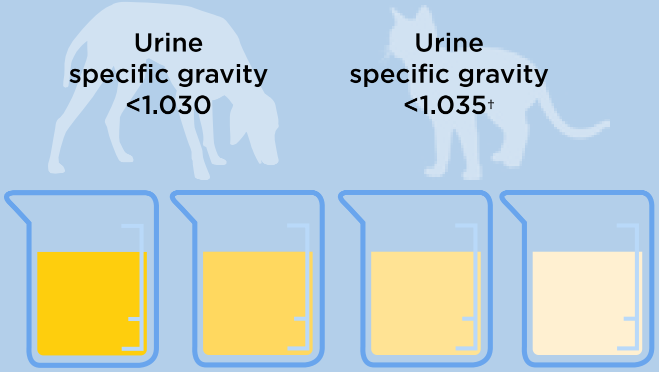
1 Increased creatinine and SDMA concentrations



Results of both tests should be interpreted in light of patient's hydration status.

plus

2



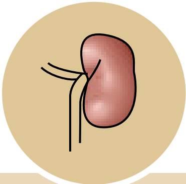
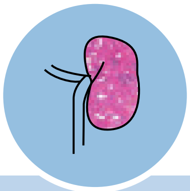
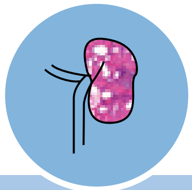
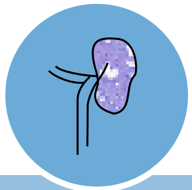
1.030	Canine	1.008
1.035	Feline	1.008

†Note that some cats can produce hypersthenuric urine in the face of renal azotemia.

Step 2: Stage CKD

Canine

Feline

		Stage 1 No azotemia (Normal creatinine)	**Stage 2** Mild azotemia (Normal or mildly elevated creatinine)	**Stage 3** Moderate azotemia	**Stage 4** Severe azotemia
Creatinine in mg/dL Stage based on	Canine	Less than **1.4** (125 µmol/L)	**1.4–2.8** (125–250 µmol/L)	**2.9–5.0** (251–440 µmol/L)	Greater than **5.0** (440 µmol/L)
	Feline	Less than **1.6** (140 µmol/L)	**1.6–2.8** (140–250 µmol/L)	**2.9–5.0** (251–440 µmol/L)	Greater than **5.0** (440 µmol/L)
SDMA* in µg/dL Stage based on stable SDMA	Canine	Less than **18**	**18–35**	**36–54**	Greater than **54**
	Feline	Less than **18**	**18–25**	**26–38**	Greater than **38**
UPC ratio Substage based on proteinuria	Canine	Nonproteinuric <0.2 Borderline proteinuric 0.2–0.5 Proteinuric >0.5			
	Feline	Nonproteinuric <0.2 Borderline proteinuric 0.2–0.4 Proteinuric >0.4			
Systolic blood pressure in mm Hg Substage based on blood pressure		Normotensive <140 Prehypertensive 140–159 Hypertensive 160–179 Severely hypertensive ≥ 180			

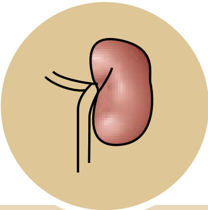
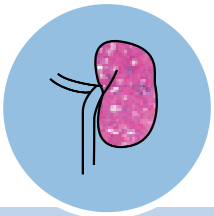
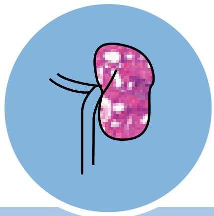
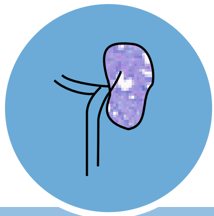
Note: In the case of staging discrepancy between creatinine and SDMA, consider patient muscle mass and retesting both in 2–4 weeks. If values are persistently discordant, consider assigning the patient to the higher stage.

*SDMA = IDEXX SDMA® Test

See www.iris-kidney.com for more detailed staging, therapeutic, and management guidelines.

Step 3: Treat CKD

Treatment recommendations

Stage 1	Stage 2	Stage 3	Stage 4
Use nephrotoxic drugs with caution	Same as Stage 1	Same as Stage 2	Same as Stage 3
Correct prerenal and postrenal abnormalities	Renal therapeutic diet	Keep phosphorus <5.0 mg/dL (<1.6 mmol/L)	Keep phosphorus <6.0 mg/dL (<1.9 mmol/L)
Fresh water available at all times	Treat hypokalemia in cats	Treat metabolic acidosis	Consider feeding tube for nutritional and hydration support and ease of medicating
Monitor trends in creatinine and SDMA to document stability or progression	Treat inappetence and nausea if present	Consider treatment of anemia	
Investigate for and treat underlying disease and/or complications		Treat vomiting, inappetence, and nausea	
Treat hypertension if systolic blood pressure persistently >160 or evidence of end-organ damage		Increased enteral or subcutaneous fluids may be required to maintain hydration	
Treat persistent proteinuria with renal therapeutic diet and medication (UPC >0.5 in dogs; UPC >0.4 in cats)			
Keep phosphorus <4.6 mg/dL (<1.5 mmol/L)			
If required, use renal therapeutic diet plus phosphate binder			

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